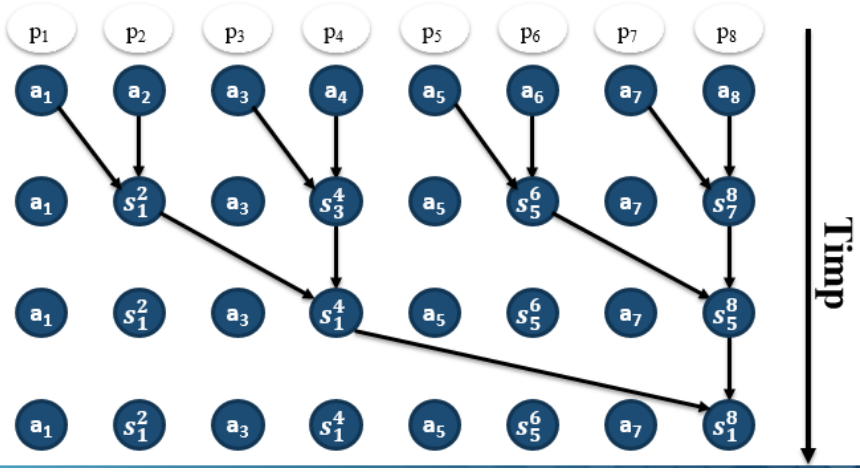
Suma elementelor unui vector

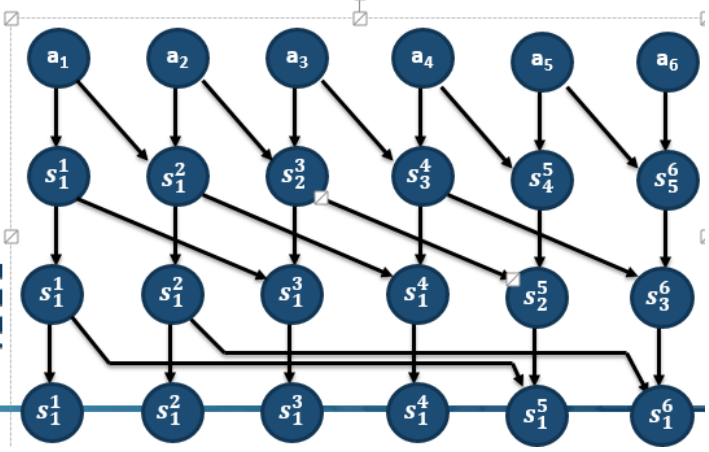
```
int a[1:n];
process suma[k=1 to n] { // suma proceselor p din figu
      for [j = 1 to sup(log2 n)] {
            if (k mod 2^j == 0)
                  a[k] = a[k-2^(j-1)] + a[k];
            barrier;
      }
}
```



Sume prefix – varianta 1

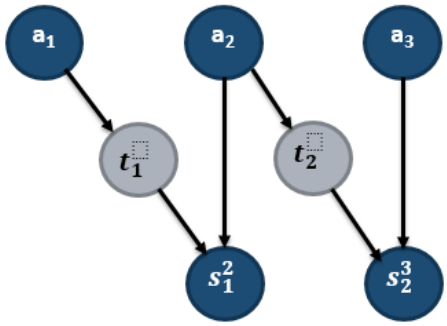
```
int a[1:n];
process suma[k=1 to n] {
      for [j = 1 to sup(log2 n)] {
            if (k - 2^(j-1) >= 1)
                  a[k] = a[k-2^(j-1)] + a[k];
            barrier;
      }
}
```

Eroare de sincronizare...



Sume prefix – varianta 2

```
int a[1:n], temp[1:n];
process suma[k=1 to n] {
      for [j = 1 to sup(log2 n)] {
            temp[k] = a[k];
            barrier;
            if (k - 2^(j-1) >= 1)
                  a[k] = temp[k-2^(j-1)] + a[k];
            barrier;
      }
}
```



Sume prefix – varianta 3

```
int a[1:n], temp[1:n];
process suma[k:1 to n] {
    int d = 1;
    while (d < n) {
      temp[k] = a[k];
      barrier;
      if (k - d >= 1) a[k] = temp[k-d] + a[k];
      barrier;
      d = 2 * d;
    }
}
```